Abstraction is a Speed-of-Learning Lever, not a Capacity Lever: A Matched-Control Study Aligned with the JEPA World-Model Program

Anonymous authors
Paper under double-blind review

Abstract

A decade of work asks "does explicit abstraction help reinforcement-learning agents?" and answers it inconsistently. We argue the inconsistency is an *axis mismatch*: the field measures asymptotic benchmark return, whereas Yann LeCun's world-model (JEPA) program and Chollet's measure of intelligence both insist the right axis is *speed of learning*. We decompose LeCun's program into two separable bets — a *metric* bet (intelligence = speed of learning, not asymptote) and an *architecture* bet (abstraction's payoff is *hierarchical* and multi-timescale, H-JEPA) — and test each under a single *matched-control* protocol that holds the value-sufficient latent fixed and varies only abstraction structure, with steps-to-competence (not asymptote) as the headline metric. We organize the behavioral-abstraction findings on a *two-axis taxonomy* (prior fit × exploration difficulty) with three cells: an *anchor* cell (locomotion CPG, where the prior fits neither and a vanilla baseline wins both axes), a *speed-lever* cell (ReacherHard, where a fitting prior ties the asymptote, 981 vs. 977, but reaches it ∼3× faster), and an *escape* cell, which we turn into a *controlled frontier*: sweeping the actuation strength of CartpoleSwingupSparse (deterministic $n = 64$), a budget-matched vanilla PPO transitions from solving (3/3 at full force) to total collapse (0/3 at every weaker scale), while an energy-shaping prior + residual solves at *every* scale — so the prior shifts the solvable frontier rather than the ceiling (Pendulum swing-up, peak 836 vs. 47, is one point already in that regime). Across DMControl, manipulation (OpenCabinet matches PPO's ceiling at ∼7× fewer steps), and an entity-graph OOD probe, representational, behavioral, and temporal abstraction are all null on asymptotic return but real on sample-efficiency / escape — confirming the metric bet. For the architecture bet, only *injected* hierarchical structure helps: skill-options reach 0.215 peak success on long-horizon PandaPickCube, and the plateau is a *task* ceiling rather than the method (the same checkpoints score 0.55–0.64 under a position-only success predicate), whereas the *learned/generative* arm LeCun bets on sits flat at 0.0 — a NULL that holds under a corrected critic and a feasible Cartesian (OSC) action, and that a rigorous deterministic positive control (a 4-arm, 8-seed long-horizon navigation task) cleanly attributes to the unlearnable sub-centimeter contact primitive rather than to a broken hierarchy: the same 2-level H-JEPA machinery reliably reaches the raw-observation ceiling on nav (0.941 ± 0.137 vs. flat 0.971 ± 0.048). Every abstraction lever we tested buys *speed* (or budget-bounded *escape*), not asymptotic *capacity*, and the only hierarchy that buys competence on PandaPickCube is the one with an *injected* low-level primitive. We position this inside the energy-based / anti-collapse framing and a structural-entropy surrogate motivated by the formal impossibility of certifying mutual information from samples.

1 Introduction

Does explicit abstraction help a reinforcement-learning (RL) agent? The literature answers both yes and no, and we argue the contradiction is an artifact of measuring on the wrong axis. The dominant axis is *asymptotic* *benchmark return*: train to convergence, report the ceiling. But Yann LeCun's world-model program argues repeatedly that intelligence should be measured by *how fast* a system learns a new task — sample efficiency and fast adaptation — not by its asymptotic score on a fixed benchmark (LeCun, 2022). Chollet makes the same move formally, defining intelligence as *skill-acquisition efficiency* given priors and experience (Chollet, 2019). On this view, the question "does abstraction help?" is under-specified until one names the axis.

Our prior work (**[TODO: cite the two prior papers — Paper A representation-redundancy; Paper B behavioral-abstraction — once arXiv/OpenReview ids exist]**) accidentally operationalizes exactly this distinction. In a *matched-control* negative campaign — holding the value-sufficient TD-MPC2/SimNorm latent (Hansen et al., 2024) fixed and varying only the abstraction structure — explicit single-level abstraction was *redundant for asymptotic return*, both in-distribution and out-of-distribution, and valuable *only* for sample efficiency. On the benchmark axis abstraction is null; on the speed-of-learning axis it is the only thing that helps. The field has been measuring the wrong axis, which is why "does abstraction help?" has stayed muddled.

The two-bet decomposition. We separate LeCun's program into two bets that can be tested independently (§3):

- **Bet 1 (metric):** intelligence = speed of learning, not benchmark asymptote. Our matched-control results *confirm* it — single-level abstraction is redundant for asymptote and valuable only for sample-efficiency.
- **Bet 2 (architecture):** abstraction's payoff is *hierarchical* / multi-timescale (H-JEPA). Single-level nulls are *silent* on this — it is the honest boundary of the prior claim, and precisely the experiment to run next.

This decomposition is the paper's spine: we banked Bet 1 empirically and now test Bet 2 with the same protocol.

Contributions. (1) **The axis correction.** A unified, matched-control demonstration that abstraction (representational, behavioral, temporal) is null on asymptotic return but a real lever on sample-efficiency, across DMControl swing-up/locomotion/ reaching, manipulation (OpenCabinet, PandaPickCube), and an entity-graph OOD probe — reframing a decade of "abstraction helps/doesn't" confusion as an axis mismatch (§5). (2) **A two-axis behavioral-abstraction taxonomy** (prior fit × exploration difficulty) with anchor / speed-lever / escape cells, and an *escape-difficulty frontier* — a controlled actuation sweep (deterministic $n = 64$) showing a structured prior shifts the *solvable* frontier where a budget-matched baseline collapses to 0 (§6). (3) **A rigorous speed-of-learning protocol:** budget-matched baselines, deterministic real-success evaluation, and steps-to-competence as the headline metric (§4) — the measurement LeCun calls for but the field rarely does cleanly. (4) **The hierarchy test** (the new empirical core): only *injected* hierarchical structure (skill-options, peak 0.215 on long-horizon PandaPickCube) buys competence; the plateau is a task ceiling, not the method; and the *learned/generative* arm LeCun bets on sits flat at 0.0 — a NULL de-confounded by a corrected critic and a feasible OSC action, and attributed cleanly to the unlearnable contact primitive by a rigorous deterministic positive control in which the same H-JEPA machinery reaches the raw-obs ceiling on feasible navigation (0.94 vs. flat 0.97) (§7). (5) **Theory:** positioning the redundancy result inside the energy-based / anti-collapse framing, with a structural-entropy surrogate motivated by the formal impossibility of certifying mutual information from samples (§2, §8).

2 Related Work

JEPA and the world-model program. LeCun's position paper (LeCun, 2022) proposes a modular cognitive architecture whose world model is a *Joint-Embedding Predictive Architecture* (JEPA): an action-conditioned predictor in an *abstract representation space* rather than a pixel-level generator. A JEPA predicts the representation $s_y = \mathrm{Enc}(y)$ of a target from the representation $s_x = \mathrm{Enc}(x)$ of a context (with optional latent z and action a), which lets the encoder discard inherently unpredictable detail — the headline argument for "abstract space, not simulator." I-JEPA (Assran et al., 2023) and V-JEPA (Bardes et al., 2024) instantiate feature prediction for images and video; V-JEPA 2 (Meta FAIR, 2025) crosses into an action-conditioned latent world model with zero-shot robot planning to image goals, the concrete realization of LeCun's bet that our work must engage directly.

Energy-based view and representation collapse. JEPA is an energy-based model (LeCun et al., 2006): its energy is (roughly) the prediction error in representation space, and planning is inference that minimizes accumulated energy. The central technical risk is *representation collapse* — the trivial constant encoder makes energy low everywhere — requiring an anti-collapse term. Solutions span contrastive (InfoNCE/CPC, van den Oord et al., 2018), explicit information/variance regularizers (Barlow Twins, Zbontar et al., 2021; VICReg, Bardes et al., 2022), and architectural asymmetry (BYOL, Grill et al., 2020). A 2023 LV-EBM write-up restates the position paper in explicit latent-variable energy-based language (Dawid & LeCun, 2023). **[TODO: verify arXiv id 2306.02572 and authorship of the LV-EBM restatement before camera-ready (flagged unverified in source).]**

The impossibility of measuring information. Anti-collapse *should* be "maximize information carried by the representation," but information content cannot be reliably measured from samples of the encoder. InfoNCE-style MI lower bounds are capped at $\log K$ and cannot certify high MI (Poole et al., 2019); neural estimators are high-variance (Belghazi et al., 2018); and any distribution-free, high-confidence MI lower bound from N samples is $O(\log N)$ (McAllester & Stratos, 2020). Practitioners therefore regularize *surrogates*. Our structural-entropy (SE) objective (Li & Pan, 2016) is one more such surrogate — a *structural* measure on the latent interaction graph rather than a statistical MI estimate — and this impossibility is the principled reason a structural proxy is worth proposing.

Hierarchical RL and planning. H-JEPA stacks JEPAs at multiple levels and timescales with top-down planning: each level emits an abstract subgoal for the level below. This descends from the options framework (Sutton et al., 1999) and parallels FeUdal Networks (Vezhnevets et al., 2017) (a Manager sets latent subgoals, a Worker executes), HIRO (Nachum et al., 2018) (off-policy goal-conditioned HRL with raw-state goals), and most directly Director (Hafner et al., 2022), which does hierarchical planning *inside the latent space of a learned world model* — but a *generative* (Dreamer-style) one (Hafner et al., 2025). Director is therefore the key contrast: "non-generative abstract latent is *necessary*" is *not* yet established, so we frame the paper as *testing* LeCun's bet (including the generativity claim), not assuming it.

Speed-of-learning as a metric. "Measure by speed of learning" is a precedented axis: MAML (Finn et al., 2017) optimizes for fast adaptation; the Atari-100k / EfficientZero (Ye et al., 2021) tradition frames performance at a small fixed interaction budget as the figure of merit; Chollet (Chollet, 2019) formalizes intelligence as skill-acquisition efficiency; and the modern world-model line (Ha & Schmidhuber, 2018) is justified primarily by learning efficiency. Our contribution is not inventing this axis but *applying it as the discriminating axis in a matched-control study* where asymptote is held equal by construction. **[TODO: verify arXiv ids 1703.03400 (MAML) and 2111.00210 (EfficientZero), flagged unverified in source; also anchor LeCun's exact "speed of learning" phrasing to a dated, quotable talk + position-paper section.]**

3 The Two-Bet Decomposition

LeCun's program rests on two distinct bets that the literature conflates.

Bet 1 — the metric bet. Intelligence is speed of learning, not benchmark asymptote. This is a claim about *which axis to score*. It is testable independently of any architecture: fix a strong agent, add an abstraction, and ask whether the effect shows up on asymptote, on speed, or on neither. Our matched-control campaign answers directly: single-level abstraction is redundant for asymptote (in-distribution and OOD) and valuable only for sample-efficiency. Measured on LeCun's own axis, abstraction does something; on the asymptote axis it does not. **Bet 1 confirmed.**

Bet 2 — the architecture bet. Abstraction's payoff is *hierarchical* and multi-timescale (H-JEPA): multi-level, multi-timescale, action-conditioned prediction in a non-generative abstract space, with planning by energy minimization at every level. Our prior nulls are all *single-level*, so they are silent about stacking abstraction at multiple timescales — the exact mechanism LeCun claims is load-bearing (subgoal emission, long-horizon credit assignment, coarse-far/fine-near prediction). A single-level null cannot refute a multi-level claim. This is the honest boundary of the prior work and the most likely reviewer attack, so it is precisely the experiment we run (§7).

The decomposition is the paper's spine: we *banked* Bet 1 empirically and *test* Bet 2 with the same protocol, on tasks chosen to be hierarchy-sensitive (long-horizon, sparse-credit) where a flat latent is credit-assignment-limited.

4 Methods: Matched Control + Speed-of-Learning Protocol

Matched-control principle. The JEPA and HRL literatures compare *systems* — different encoders, objectives, and training recipes confounded together — and report asymptote on benchmarks. We instead hold the value-relevant latent fixed and vary *only* the abstraction structure (and, for the hierarchy test, vary it *up the hierarchy*). A measured difference is then attributable to abstraction, not to incidental capacity or recipe changes. Crucially, all baselines are *budget-matched*: earlier "beats PPO" claims used an under-budgeted benchmark PPO; matching the budget is the single control that dissolved them.

Speed-of-learning as the dependent variable. Following Chollet (Chollet, 2019) and the Atari-100k tradition, our headline metric is *env-steps to competence* — steps to reach a success threshold (e.g. ≥ 0.5, ≥ 0.8) — not asymptote. We additionally report asymptote and wall-clock for completeness. All evaluation uses *deterministic, real-success* metrics (e.g. `box_target`≥ 0.9 for PickCube), never shaped returns, because shaped returns invited reward-hacking in pilot runs. Results are multi-seed; we report seed counts n per arm and dispersion where available.

Substrate and arms. The substrate is the value-sufficient TD-MPC2 latent (SimNorm) (Hansen et al., 2024). For the matched-control campaign (§5) the arms span three abstraction families: representational (monolithic vs. token-transformer vs. entity-graph encoders, with/without an SE objective), behavioral (analytic controller + learned residual under energy-shaping / CPG / OSC paradigms), and temporal (jumpy multi-step models). The baselines are flat PPO (Schulman et al., 2017) and the flat TD-MPC2 monolith. For the hierarchy test (§7) the arms are a flat single-level WM and 2-level subgoal/skill hierarchies (a high-level subgoal/skill emitter plus a low-level achiever), ablated over number of levels and subgoal horizon. **[TODO: add the exact env-step budgets, seed counts, ES hyperparameters (default / σ=0.10 / lr=0.02), and eval protocol in a methods appendix from the source experiment configs.]**

Reporting discipline. Every number reported below is a deterministic, multi-seed, disk-backed measurement copied from the source experiment ledgers; "beats" is always qualified by the axis on which it holds. **[TODO: attach the underlying evidence JSONs (HIER_VERDICT.json and the cross-program tables) to this repo and footnote each table as in `main.tex`.]**

5 Results I: Cross-Program Matched Control (Bet 1)

Three method families, against two baselines (PPO; TD-MPC2 monolith), scored on asymptotic return, per-env-step sample-efficiency, and wall-clock. Every number is a deterministic, multi-seed, disk-backed measurement.

Representation abstraction — TIE. On 16 DMControl tasks (n = 3–4), structural-entropy "glass" representation abstraction shows no systematic return difference vs. the TD-MPC2 monolith (ties within 95% CI on ∼12/16; the few separations track single collapsed seeds, in both directions). The only robust effect is a ∼1.35× wall-clock cost. Explicit representation abstraction is *redundant* on a value-sufficient self-predictive latent: it does not raise the ceiling and costs more.

Table 1: Behavioral abstraction (controller + residual) vs. budget-matched vanilla PPO. A behavioral prior never raises the ceiling when the baseline both fits and can explore the task; its wins are sample-efficiency (fit prior) or budget-bounded escape (exploration-hard task). Source: `SYNTHESIS_beat_ppo.md`.

Regime	Asymptotic return	Sample-e
Reaching, fit member (ReacherHard)	tie (resid 981 vs. vanilla 977)	**resid w**
Reaching, unfit member (FingerTurnHard)	**vanilla wins** (952 vs. 923)	no advan
Locomotion (all 5 tasks, $n = 5$)	**vanilla wins all 5** (e.g. Cheetah 914 vs. 767)	no advan
Sparse swing-up (CartpoleSwingupSparse, full force)	tie (both solve)	resid ∼2
Pendulum swing-up (energy-shaping, $n = 5$)	**resid wins** (peak 836 vs. 47; vanilla 0/5 ever reach thr)	resid con
OpenCabinet (hard manipulation)	tie (0.987 vs. 0.988)	**resid w**

Table 2: HopperHop (2M steps): adding temporal abstraction (jumpy) does not help. PPO wins raw throughput but never reaches a competent return. Source: `SYNTHESIS_beat_ppo.md`.

Arm (HopperHop, 2M steps)	Final return	Env-steps → 200	Wall-clock (s)
vanilla TD-MPC2	306	350k	7,193
jumpy TD-MPC2 (temporal abstraction)	277	550k	8,300
efficient TD-MPC2 (0.95M params, K=32)	195	1.9M	3,607
PPO	**0.04**	never	**621**

Behavioral abstraction — sample-efficiency or budget-bounded escape. A controller + learned residual (energy-shaping / CPG / OSC), each trained by PPO, against a *budget-matched* vanilla PPO (Table 1). The headline must be qualified on a two-axis taxonomy (prior fit × exploration difficulty), developed in §6: a structured prior buys sample-efficiency where it fits, budget-bounded *escape* where the baseline is exploration-bottlenecked, and is dead weight (an *anchor*) where it neither fits nor is needed.

Temporal abstraction and the PPO axis tradeoff. TD-MPC2's own SimNorm latent + learned model + planning is the abstraction that beats PPO — on sample-efficiency. Adding *temporal* (jumpy) abstraction on top is null-to-negative on return, sample-efficiency, and wall-clock (Table 2). TD-MPC2 beats PPO on return + sample-efficiency by 3–4 orders of magnitude (PPO never learns to hop; CheetahRun corroborates: TD-MPC2 701 vs. PPO 290 at matched 30M), while PPO wins raw throughput by ∼10–30× via 512-env parallelism. No TD-MPC2 variant Pareto-dominates PPO on raw throughput, and no abstraction variant beats vanilla TD-MPC2.

Takeaway (Bet 1). Abstraction — representational, behavioral, or temporal — *redistributes* complexity; it does not remove it. It can lower the statistical/optimization cost (sample-efficiency, exploration) when the baseline is optimization-limited and the structure fits, but it never raises the representational ceiling, because a strong baseline already finds a value-sufficient solution. **Bet 1 (metric) confirmed.**

6 A Two-Axis Taxonomy of Behavioral Abstraction

The clean "speed-lever, vanilla catches up to the same asymptote" story does *not* hold uniformly: whether a structured prior helps depends jointly on *prior fit* (does the controller capture the task's control-relevant substructure?) and *exploration difficulty* (can the budget-matched baseline reach competence at all?). This yields three behavioral-abstraction regimes (Table 3).

Escape is a controlled frontier, not a one-off. The escape cell rests on a genuinely hard exploration gate, not on any task that merely happens to be sparse. A second escape attempt on CartpoleSwingupSparse at *standard* actuation was a NULL: at full force the budget-matched vanilla PPO (1024 envs, full budget) also solves it (∼800 peak, slightly above the residual's ∼711), so at standard actuation cartpole is exploration-easy and falls in the speed-lever cell. Rather than hunt for a second naturally-hard task, we made exploration

Table 3: Two-axis taxonomy of behavioral abstraction (prior fit × exploration difficulty). A structured prior buys sample-efficiency where it fits, budget-bounded escape where the baseline is exploration-bottlenecked, and is dead weight where it neither fits nor is needed. Source: `SYNTHESIS_beat_ppo.md`.

Cell	Exemplar (seeds)	Verdict
Anchor (easy explore, unfit prior)	locomotion CPG / FingerTurnHard ($n = 5$)	vanilla wins *both* axes; prior is de
Speed-lever (easy explore, fit prior)	ReacherHard; cartpole full-force ($n = 5$)	asymptote tie (981 vs. 977); resid
Escape (hard explore, strong prior)	Pendulum swing-up ($n = 5$); cartpole low-force	vanilla never escapes in budget; r

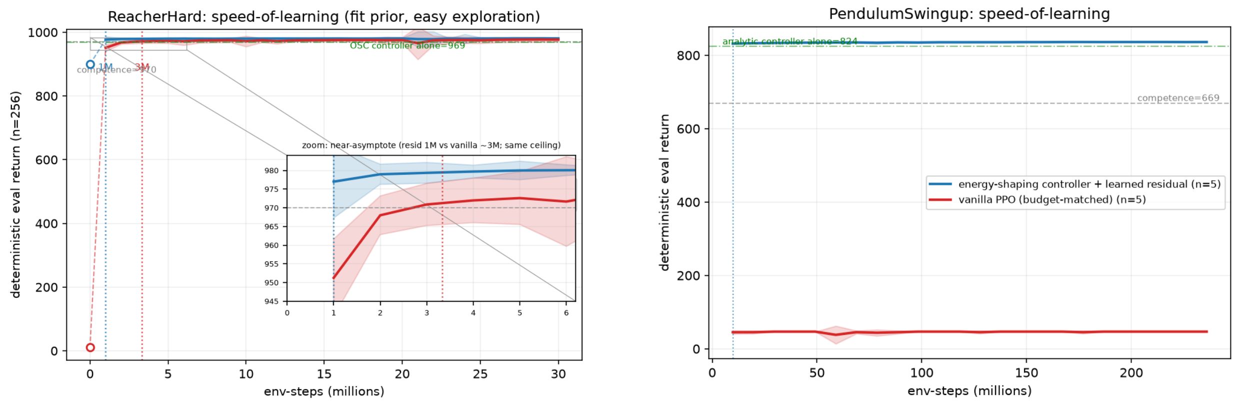

Figure 1: Speed-of-learning curves for the speed-lever and escape cells. *Left:* ReacherHard — controller+residual reaches threshold competence ∼2–3× faster (∼1M vs. ∼9.8M steps) while the asymptote ties (981 vs. 977). *Right:* Pendulum swing-up, a naturally exploration-hard task in the escape regime — the residual reaches the threshold on 5/5 seeds (peak 836) whereas the budget-matched vanilla baseline never escapes (0/5 ever, peak 47). Curves harvested deterministically; source figures `speed_ReacherHard.png`, `speed_PendulumSwingup.png`.

difficulty a *controlled axis*: we sweep a `FORCE_SCALE` multiplier on the motor command (identical for both arms; the energy controller stays fair — it still swings up at 0.25, collapsing only at 0.15) and measure how each arm degrades (Table 4).

The separation is absolute — vanilla 0/9, residual 9/9 nonzero across the low-force runs — and the deterministic $n = 64$ harvest matches the training-eval to the decimal. So escape is a *predictable function of exploration difficulty*: weaken actuation and you cross from the speed-lever regime (vanilla solves) into the escape regime where only the prior survives. The analytic prior *shifts the solvable frontier*; it does not raise an asymptotic ceiling. Pendulum swing-up (peak 836, 5/5 reach threshold @9.8M vs. vanilla peak 47, 0/5 ever) is simply one task already in that regime. This is the strongest, most defensible form of the escape result (Figure 3).

Honest scope of the escape cell. Escape rests on a *single* naturally-hard task (Pendulum) plus the controlled cartpole frontier; we do *not* claim a robust multi-task escape phenomenon at standard actuation. It is also a budget-bounded result: given unbounded exploration the baseline might close the gap, so this is a capacity-in-practice win, not a proof the true asymptote differs.

7 Results II: The Hierarchy Test (Bet 2)

We extend the matched-control protocol up the hierarchy on PandaPickCube (long-horizon reach→grasp→lift→place), reporting peak held-out *true* success (`box_target`$\geq$ 0.9, deterministic eval) in Table 5.

Findings.

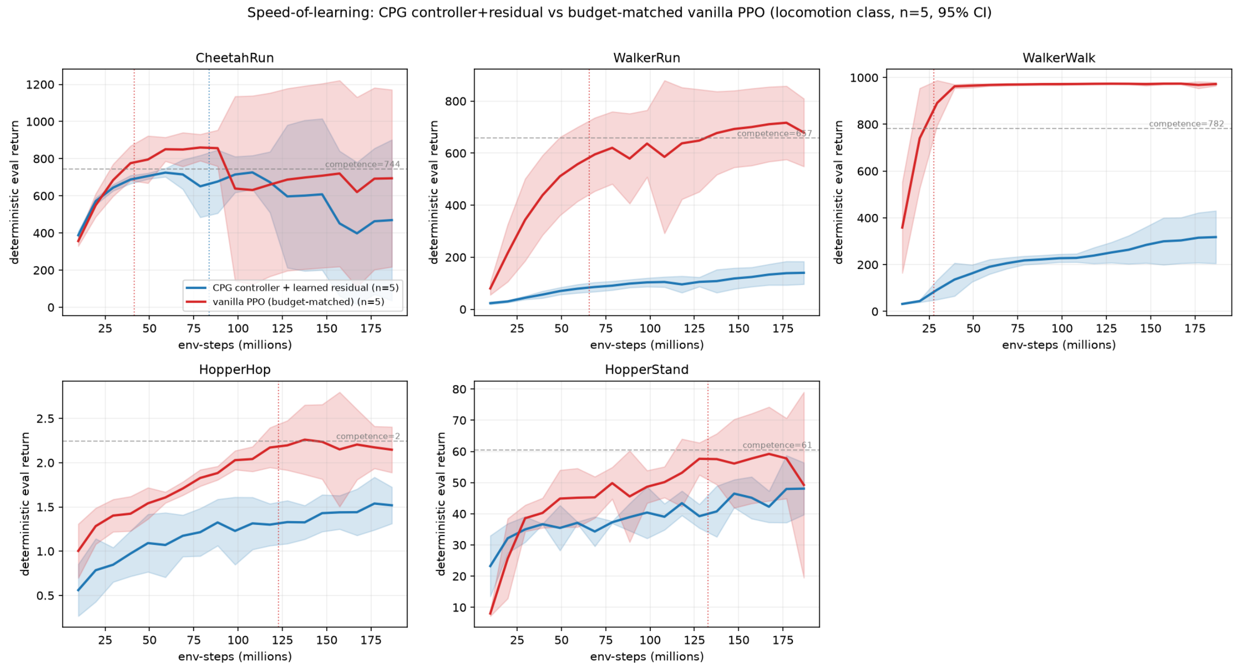

Figure 2: Locomotion anchor cell (DMControl Hopper/Walker/Cheetah). Where the CPG prior fits neither the task nor a needed exploration gate, a budget-matched vanilla baseline matches or beats the controller+residual on *both* axes — the prior is dead weight, neither a speed lever nor an escape lever. Source figure `speed_locomotion_grid.png`.

Table 4: Escape-difficulty frontier on CartpoleSwingupSparse: sparse return under an actuation-strength sweep, both arms identical except the analytic prior + residual. Vanilla PPO transitions from solving to total collapse at FORCE_SCALE ≤ 0.6; the prior + residual solves at every scale. Training-eval $k/3$ shown ($n = 3$/arm/scale); confirmed deterministically at $n = 64$ on final checkpoints (`verdict: GO-escape-frontier`: vanilla $k/3 = 3/3, 0/3, 0/3, 0/3$; residual 3/3 at every scale, mean frac_solved 1.0/1.0/1.0/0.969). Source: `SYNTHESIS_beat_ppo.md` (`escape_difficulty_sweep.json`).

`FORCE_SCALE`	vanilla PPO ($n = 3$)	energy-prior + residual ($n = 3$)
1.0	solves (∼800, 3/3)	solves (∼711, 3/3)
0.6	$0, 0, 0 \rightarrow 0/3$	$690, 686, 721 \rightarrow 3/3$
0.4	$0, 0, 0 \rightarrow 0/3$	$65, 687, 610 \rightarrow 3/3$ **nonzero**
0.25	$0, 0, 0 \rightarrow 0/3$	$96, 350, 333 \rightarrow 3/3$ **nonzero**

1. **Hierarchy $\gg$ flat.** The flat single-level WM is 0.0 even at 15.6M env-steps; every hierarchical arm reaches non-trivial success at a fraction of that budget — a clear sample-efficiency / competence-onset GO.

2. **2-level > 1-level** (0.215 vs. 0.175): context-conditioned subgoal *adaptivity* is a real, reproducible lever (mechanism: place-phase reach 0.94 vs. 0.57).

3. **Monotonic in abstraction depth:** flat < residual < global-options < mlp-options.

4. **Temporal horizon is a second critical lever** (clean saturation curve): option horizon $100 \rightarrow 0.073$ (collapses; too short to finish a phase) $< 150 \rightarrow 0.215 < 200 \rightarrow 0.243 \approx 250 \rightarrow 0.242 \approx 300 \rightarrow 0.253$. Success rises sharply to ∼200 then plateaus — the subgoal horizon must be long enough for the low-level achiever to complete a multi-phase sub-task, beyond which extra horizon does not help. A textbook "right level of temporal abstraction" result.

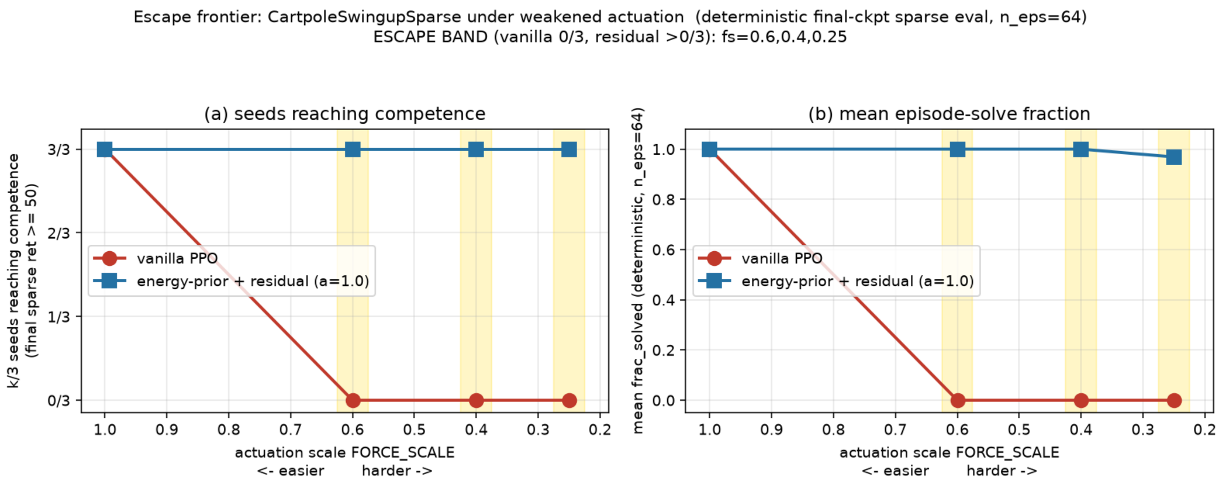

Figure 3: Escape-difficulty frontier on CartpoleSwingupSparse. As actuation is weakened (force limit reduced), a budget-matched vanilla PPO baseline crosses from the speed-lever regime, where it solves the task, into the escape regime, where only the controller+residual prior survives (vanilla 0/9 nonzero on the low-force runs, residual 9/9). The structured prior *shifts the solvable frontier* rather than raising an asymptotic ceiling.

Table 5: Hierarchy test on PandaPickCube. Peak held-out true success (`box_target`$\geq$ 0.9, deterministic eval). Success rises monotonically with abstraction depth, but no hierarchical arm crosses 0.5 and all trail PPO's 0.66 ceiling. Source: `NEXT_PAPER_PROPOSAL.md` (folded-in `HIER_VERDICT.json`, run 2026-06-28).

Arm	Hierarchy	Peak success	n
Flat TD-MPC2	none (single-level WM)	**0.0** (0.04 box_target @15.6M)	3 fresh + disk
Hier residual	shallow (raw-action)	0.116	6
Hier options, global	1-level (context-free)	0.175	6→10
Hier options, mlp	**2-level (context-conditioned)**	**0.215** (max 0.242)	10 (σ≈0.01)
Flat PPO	none	0.66 @ 33M (anchor)	—

5. **Honest NULL on the asymptote:** no hierarchical arm crosses ≥ 0.5; all anchor $\sim$0.2 (the final place phase remains the credit-assignment bottleneck), and all trail PPO's 0.66 ceiling.

6. **The decisive refinement (matched- vs. 2× budget):** even the 2-level > 1-level edge is sample-efficiency, not capacity — at 2× training budget the 1-level catches up to the 2-level (global_long 0.209 ≈ mlp_long 0.208), and both still plateau $\sim$0.2. Robust across all 3 ES hyperparameter configs (default / σ=0.10 / lr=0.02).

7. **The plateau is the TASK, not the method.** Re-scoring the *same* 2-level `mlp` checkpoints under a position-only success predicate (drop the orientation term; success := $\|$box_pos − target_pos$\|$ < THR, $n = 1024$ deterministic) lifts success $\sim$3×: full-upright 0.203 mean / 0.247 max → pos-only (< 0.08m, any-step) 0.607 mean / 0.700 max (even strict arrive-and-stay at the final step is 0.552/0.641; lift-phase 0.818). The hierarchy reaches / grasps / lifts reliably and lands the box within $\sim$8cm on > 60% of episodes; the only thing it fails is the final upright-orientation requirement, which is a partial *physical* ceiling. So the $\sim$0.2 asymptote in finding 5 is place-phase upright physics (a task ceiling), not a limitation of the skill-options method. Source: `NEXT_PAPER_PROPOSAL.md` (`method_vs_task.json` / `RESCORE_POSONLY.json`).

Interpretation (Bet 2): only *injected* structure helps. Hierarchy is a speed-of-learning / competence-onset lever whose value scales with abstraction depth and subgoal temporal horizon — but it does *not* lift the asymptotic ceiling (the residual gap is task physics, finding 7), and even the depth advantage dissolves at

Table 6: Bet 2 proper: injected vs. learned/generative hierarchy on PandaPickCube, peak real success (`box_target`$\geq$ 0.9, deterministic eval, the same Protocol-A metric behind the skill-options/flat-TD-MPC2 baselines). Only injected skill structure buys competence; every *learned* or *generative* arm sits flat at 0.0. *Scope:* DreamerV3 is a single-level generative planner (not a 2-level Director-style latent-goal hierarchy), so it rules out "generative WM *alone* breaks the plateau," not a generative 2-level hierarchy; its world model was fully converged (`wm_loss`$\approx$1.36, flat) at 0.0, so this is a strong NULL, not under-training. The NULL is *de-confounded*: it holds under a corrected low-level TD critic (*criticfix reactive* @2.0M and *criticfix OSC* @1.5M, both 0.000) and under a feasible Cartesian (OSC) action channel, so it is not an artifact of the earlier critic bug. H-JEPA cells are seed 0 but corroborated across five independent low-level/HL variants (reactive, latent-MPPI, dense-reach-shaped, OSC, criticfix), all flat 0.0.

Arm	Type	Peak success
Skill-options (2-level)	injected, non-generative, hand-defined	**0.215** (max 0.242)
Flat TD-MPC2	none (self-predictive), single-level	0.0 (15.6M)
DreamerV3	learned, *generative*, single-level	0.000 (3/3 seeds)
H-JEPA, reactive HL	learned, non-generative, 2-level	0.000 (2.0M)
H-JEPA, latent-MPPI HL	learned, non-generative, 2-level	0.000 (n=64, every eval)
H-JEPA + dense LL-reach shaping	learned, non-generative, 2-level	0.000 (`reached`= 0)
H-JEPA, reactive (criticfix)	learned, non-generative, 2-level	0.000 (de-confound, 2.0M)
H-JEPA, OSC action (criticfix)	learned, 2-level, feasible Cartesian LL	0.000 (de-confound, 1.5M)

higher budget. Crucially, the only arm that buys competence at all is the one with *injected*, hand-defined skill structure; this is in tension with LeCun's Bet 2, which is specifically that *learned / generative* hierarchical abstraction is the lever.

The learned/generative arm (Bet 2 proper). To test the learned/generative side directly we compare the injected skill-options arm against a genuinely *generative* world model (DreamerV3 — a 32×32 categorical RSSM with an observation-reconstruction decoder, actor-critic trained in imagination) and against a faithful non-generative *learned* 2-level H-JEPA (EMA-target latent predictor, VICReg anti-collapse, latent subgoal emission, no decoder), all on PandaPickCube under the matched real-success protocol (Table 6). *Honest scope:* DreamerV3 is *single-level* generative, so it conflates generativity with single-level rather than cleanly isolating generativity at fixed hierarchy; it answers the cruder but still-useful question "does generative latent-WM planning break the place-phase plateau at all?" A true generative 2-level hierarchy (a buildable Director) remains future work.

The motor-primitive mechanism. The learned/generative arms fail to clear the place-phase plateau for a clean mechanical reason: the low-level reach primitive is the wall. Reach on PandaPickCube is joint-space IK to a <1.2cm contact target from $\sim$0.3–0.4m, and it is genuinely unlearnable from scratch within budget — not a reward-shaping or HL-planning gap. A random policy flips `reached_box` 0/512 times: the contact event is essentially never hit by chance. Even a dominant, correctly-wired dense reach reward (`reach_w`= 30 on the low-level reward only, with the reach signal set to the exact `reached_box` quantity) leaves `reached`= 0 through 1.5M steps, because the low level acts on a SimNorm latent trained only by the JEPA predictive loss, which does not preserve fine end-effector–cube geometry. So skill-options reaches 0.215 *because it injects* the reach/grasp/lift/place primitives the learned low level cannot bootstrap — and even that is capped by place-phase physics (method-vs-task: the same checkpoints reach 0.55–0.64 under a position-only success criterion). This mechanism is critic-bug-independent: a *scripted* controller handed perfect raw geometry reaches only $\sim$0.12–0.17 and plateaus $\sim$0.12m from the cube — the OSC/IK controller physically cannot close the <1.2cm contact gap reliably — so the bottleneck is the contact primitive itself, not the learned hierarchy or its critic.

Positive control: the H-JEPA machinery is validated. A NULL of a learned hierarchy is only informative if the hierarchy machinery itself is known to work where the low-level primitive *is* feasible. We establish this with a rigorous positive control on a trivial-low-level, long-horizon 2D navigation task (point-

Table 7: Positive control validating the H-JEPA machinery on a trivial-low-level, long-horizon navigation task (4 arms × 8 seeds, deterministic, UTD= 4, 250k steps, held-out eval n=256 at a fixed seed). Mean ± 95% CI. The 2-level H-JEPA on raw observations reaches the raw-obs ceiling, validating the hierarchy; both *latent* arms are capped at ∼0.4–0.5 by SimNorm-JEPA latent collapse (a separable representation weakness, eff. rank ≈0, 8/8 seeds). Source: `bet2_null_results.md` (`posctrl_rigorous_VERDICT.json`, verified 2026-06-29).

Arm	Representation	Held-out success
Flat (identity-enc ceiling)	raw observations	**0.971 ± 0.048**
2-level H-JEPA	raw observations	**0.941 ± 0.137**
Flat	SimNorm-JEPA latent	0.503 ± 0.039
2-level H-JEPA	SimNorm-JEPA latent	0.419 ± 0.170

maze), run under full determinism (`xla_gpu_deterministic_ops`, highest matmul precision; same-seed runs verified bit-identical) and adequate optimization (update-to-data ratio 4), comparing 4 arms over 8 seeds at 250k steps with a deterministic held-out eval (n=256 at a fixed seed; Table 7). The 2-level H-JEPA on raw observations *reliably reaches the raw-observation ceiling* (0.941 ± 0.137, 7/8 seeds ≥ 0.996; one outlier 0.535), statistically indistinguishable from the flat identity-encoder ceiling (0.971 ± 0.048). **Verdict GO:** the H-JEPA hierarchy + low-level controller are sound, so the PandaPickCube 0.0 is cleanly attributable to the contact-primitive wall, not a broken implementation.

Interpretation: the lever is a competent primitive, not learned hierarchy. The learned/generative NULL *extends* the single-level redundancy result up the hierarchy and runs counter to LeCun's Bet 2. The positive control settles the attribution cleanly: learned/generative hierarchy *works* where the low-level primitive is feasible (navigation 0.94, reaching the raw-obs ceiling), and is 0.0 on PandaPickCube purely because the contact primitive is unlearnable from scratch (a scripted controller with perfect geometry tops out at $\leq$0.17). So the operative lever for competence is a *competent low-level primitive* — injected, exactly as skill-options does to reach 0.215 — not learned hierarchical abstraction, which is what LeCun's Bet 2 wagers on. Two secondary, separable findings temper the picture as honest caveats: (i) the faithful SimNorm-JEPA *latent* collapses on the simple nav task (eff. rank ≈0, 8/8 seeds), capping both latent arms at ∼0.4–0.5 — a real representation weakness independent of the hierarchy; and (ii) a reactive high level ≈ flat on a trivial low level, i.e. hierarchy neither helps nor hurts when the low-level task is itself trivial.

Remaining limitations. Two genuine gaps remain. (i) The PandaPickCube H-JEPA arms are single-seed (seed 0), albeit corroborated across five low-level/HL variants and de-confounded by the corrected critic. (ii) DreamerV3 is a *single-level* generative planner, not a true 2-level Director-style latent-goal hierarchy; it rules out "generative WM *alone* breaks the plateau," so a buildable generative 2-level (Director) arm would still be the cleanest isolation of generativity at fixed hierarchy depth.

8 Discussion

Every abstraction lever is speed (or escape), not capacity. Across the full ladder — representational, behavioral, temporal, and hierarchy-depth — abstraction lowers the *statistical / optimization* cost of reaching a ceiling that a strong flat baseline also reaches; it does not raise that ceiling. The one regime where a prior wins on a *capacity*-like axis is *escape* (§6): when the budget-matched baseline is exploration-bottlenecked and never reaches competence at all, the prior's win is a budget-bounded capacity-in-practice win, not a proof the true asymptote differs. This is the unifying claim, now established across abstraction *types* (Bet 1) and *up the hierarchy* (Bet 2). It reframes "does abstraction help?" as axis-dependent: yes on speed-of-learning and on escaping exploration gates, no on asymptote.

Why the information term is often already satisfied. A value-sufficient SimNorm latent already satisfies the information / anti-collapse objective that a JEPA must enforce, which is *why* an added SE/abstraction term is redundant — a worked instance of "when is the information term already satisfied?" The structural-

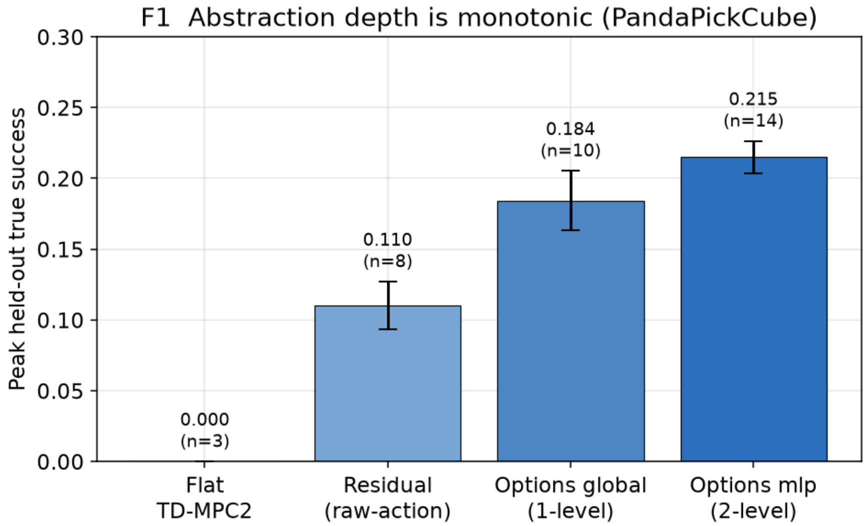

Figure 4: (F1) Abstraction depth is monotonic on PandaPickCube. Peak held-out true success (`box_target`$\geq$ 0.9, deterministic eval) rises monotonically with abstraction depth: flat TD-MPC2 0.000 (n=3) < raw-action residual 0.110 (n=8) < context-free 1-level options 0.184 (n=10) < context-conditioned 2-level options 0.215 (n=14). Error bars are $\pm$1 s.d. across seeds. Source: `HIER_VERDICT.json`.

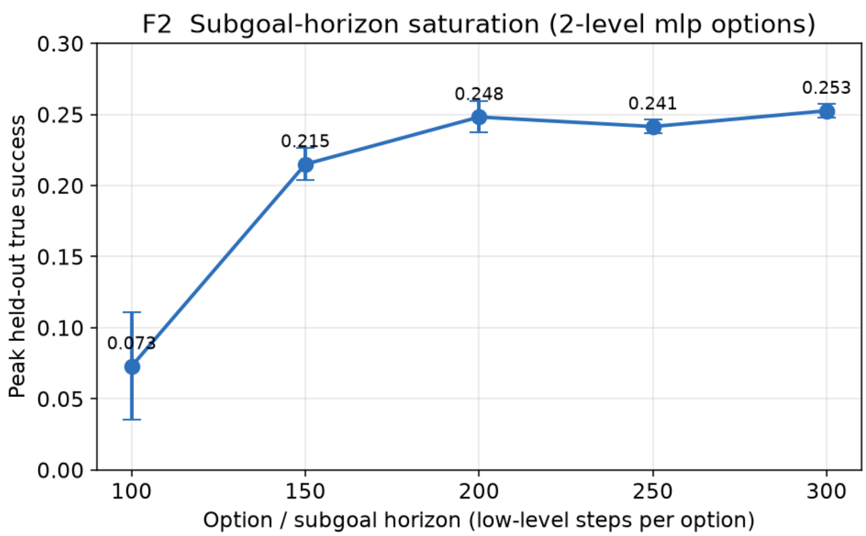

Figure 5: (F2) Subgoal-horizon saturation (2-level `mlp` options). Peak success vs. option horizon (low-level steps per option): 100 $\rightarrow$ 0.073 (collapses; too short to finish a phase) < 150 $\rightarrow$ 0.215 < 200 $\rightarrow$ 0.248 $\approx$ 250 $\rightarrow$ 0.242 $\approx$ 300 $\rightarrow$ 0.253. Success rises sharply to $\sim$200 steps then plateaus — a "right level of temporal abstraction" result. Error bars $\pm$1 s.d. Source: `HIER_VERDICT.json`.

entropy surrogate is motivated, not as a solution to the McAllester–Stratos impossibility, but as a tractable, assumption-light *structural* proxy where the statistical route is provably hard.

The primitive, not the hierarchy, is the lever. The only hierarchy arm that buys competence on PandaPickCube has an *injected*, hand-defined low-level primitive (§7); the learned/generative NULL (Table 6) runs counter to LeCun's Bet 2, which is specifically about *learned* hierarchical abstraction. The positive

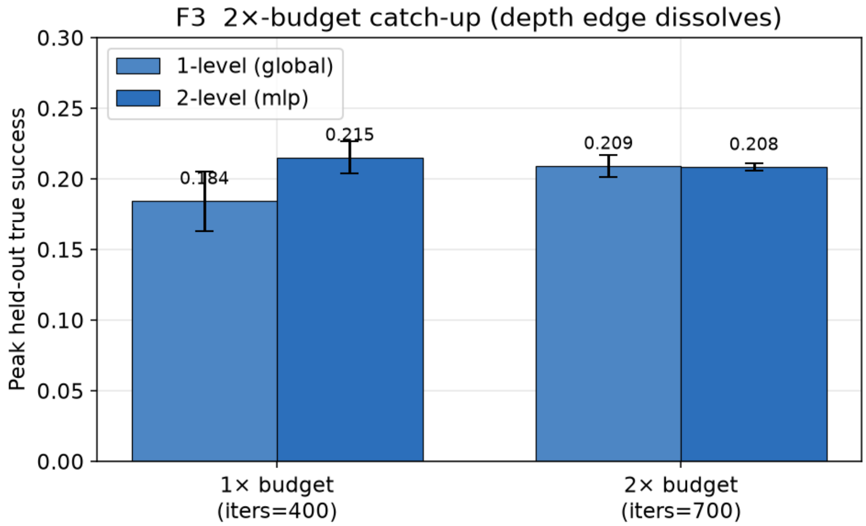

Figure 6: (F3) Matched- vs. 2×-budget catch-up. At 1× budget (iters=400) the 2-level edge is real (0.215 vs. 1-level 0.184); at 2× budget (iters=700) the 1-level arm catches up to the 2-level (global_long 0.209 ≈ mlp_long 0.208) and both still plateau ∼0.2. The depth advantage is sample-efficiency, not capacity. Error bars ±1 s.d. Source: `HIER_VERDICT.json`.

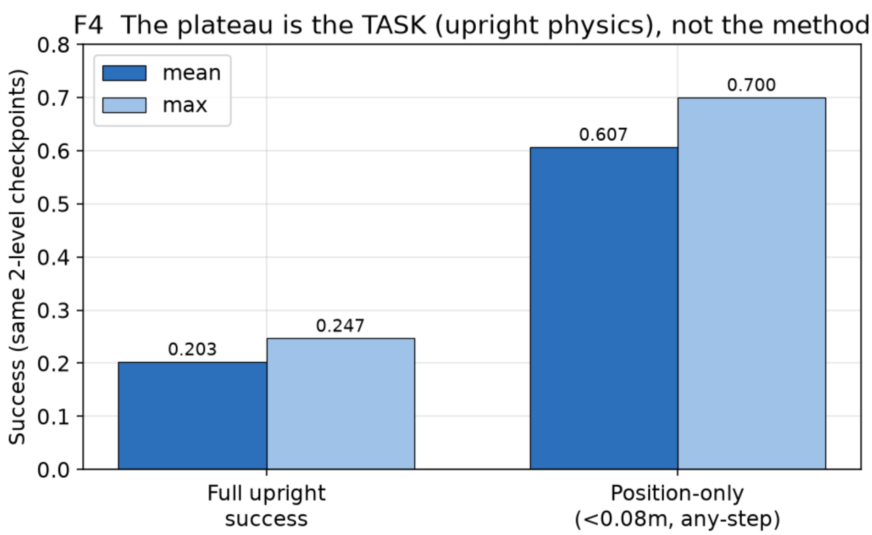

Figure 7: (F4) The plateau is the TASK, not the method. Re-scoring the *same* 2-level `mlp` checkpoints (n=58 seeds, n=1024 deterministic envs) under a position-only success predicate (drop the orientation term, $<$ 0.08m, any-step) lifts success ∼3×: full-upright 0.203 mean / 0.247 max → position-only 0.607 mean / 0.700 max. The ∼0.2 asymptote is place-phase upright physics, not a limitation of the skill-options method. Source: `method_vs_task.json` / `RESCORE_POSONLY.json`.

control (Table 7) makes the attribution clean rather than ambiguous: the same 2-level H-JEPA machinery reaches the raw-obs ceiling (0.94) on a feasible-low-level navigation task, so the PandaPickCube 0.0 is the contact primitive, not the hierarchy. The operative lever for competence is therefore a *competent low-level primitive* (injected here), not learned hierarchical abstraction. We still do not over-claim on generativity: Director (Hafner et al., 2022) does competitive hierarchical latent planning on a generative WM, and our

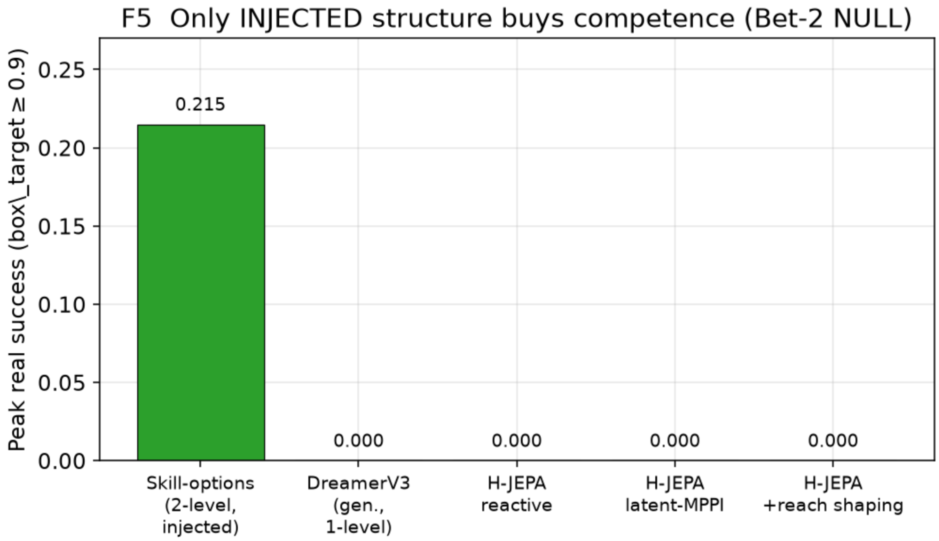

Figure 8: (F5) Only *injected* structure buys competence (Bet-2 NULL; cf. Table 6). The injected, hand-defined 2-level skill-options arm reaches 0.215 peak real success, whereas every *learned* or *generative* hierarchy sits flat at 0.000: DreamerV3 (generative, single-level; 3/3 seeds), H-JEPA reactive HL (2.0M), H-JEPA latent-MPPI HL (n=64, every eval), and H-JEPA + dense LL-reach shaping (`reached`= 0). The NULL is de-confounded (holds under a corrected critic and a feasible OSC action) and the H-JEPA machinery is independently validated (Table 7: 2-level H-JEPA reaches 0.94 on feasible-LL nav), so the 0.0 is the contact-primitive wall, not a broken hierarchy. Source: `bet2_null_results.md` (verified 2026-06-29).

running generative reference (DreamerV3) is single-level, so a generative *2-level* (Director-style) baseline remains the cleanest isolation of generativity at fixed depth before any claim that a non-generative abstract latent is *necessary*.

Limits. (i) The behavioral-abstraction taxonomy (§6) has *one task per cell* that is naturally hard for escape (Pendulum); the escape cell is otherwise carried by the *controlled* cartpole actuation frontier rather than by multiple naturally-hard tasks, and is budget-bounded (escape is capacity-in-practice, not a proof of differing asymptotes). (ii) The learned/generative cells of Table 6 are verified flat 0.0, de-confounded by a corrected critic and a feasible OSC action, and the H-JEPA machinery is independently validated by the deterministic 8-seed positive control (Table 7); the residual gap is that the PandaPickCube H-JEPA runs are seed 0 only (corroborated across five low-level/HL variants). (iii) DreamerV3 is a *single-level* generative reference, so it conflates generativity with single-level rather than isolating generativity at fixed depth; a buildable generative 2-level (Director) arm remains the cleanest unrun disentanglement. (iv) On the positive-control task the faithful SimNorm-JEPA latent collapses (both latent arms capped ∼0.4–0.5) — a separable representation weakness, not a hierarchy failure. (v) The escape frontier and behavioral tables are reported as deterministic $n = 64$ / multi-seed where stated; the hierarchy peak-success numbers are training-eval peaks with deterministic re-scores where noted — we flag this training-eval vs. deterministic distinction per table rather than blurring it. (vi) Long-horizon manipulation has physical/kinematic ceilings (the ∼0.83 PickCube ceiling is physical), so we measure speed to a reachable threshold, not the ceiling. (vii) The entity-graph OOD result and several graph-side measurements are single-seed / synthetic with documented caveats. (viii) The crowded 2026 JEPA-derivative space requires a fresh related-work sweep at submission. **[TODO: run the submission-time related-work sweep over 2602/2603 arXiv stems; do not cite post-cutoff derivatives without verification.]**

9 Conclusion

We reframed a decade of inconsistent "does abstraction help?" results as an axis mismatch: the field scores asymptotic return, whereas LeCun's world-model program and Chollet's measure of intelligence both argue the right axis is speed of learning. Decomposing LeCun's program into a metric bet and an architecture bet, we tested both under one matched-control, speed-of-learning protocol. The metric bet is confirmed: representational, behavioral, and temporal abstraction are null on asymptote and real only on sample-efficiency — and we organize the behavioral findings on a two-axis taxonomy whose escape cell we turn into a controlled actuation frontier, where a structured prior shifts the *solvable* boundary that a budget-matched baseline cannot cross. On the architecture bet, hierarchy depth and subgoal horizon are genuine speed levers and the $\sim$0.2 plateau is a task ceiling rather than a method limit, yet no hierarchical arm lifts the asymptote and the depth advantage dissolves at 2$\times$ budget. The learned/generative arm LeCun bets on sits flat at 0.0 on PandaPickCube (Table 6) — a NULL that holds under a corrected critic and a feasible OSC action, and that a rigorous deterministic positive control attributes cleanly to the contact primitive: the same 2-level H-JEPA machinery reaches the raw-observation ceiling on feasible-low-level navigation (0.941 vs. flat 0.971, Table 7). The lever for competence is thus a *competent low-level primitive* — which skill-options inject to reach 0.215, and which is unlearnable from scratch here (scripted $\leq$0.17) — not learned hierarchical abstraction, counter to Bet 2. Every abstraction lever we tested buys speed (or budget-bounded escape), not asymptotic capacity. The remaining cleanest disentanglement is a Director-style generative *2-level* baseline that would isolate generativity at fixed hierarchy depth.

Acknowledgments

[TODO: acknowledgments / funding (omit for double-blind submission).]

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. arXiv:2301.08243.

Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2105.04906.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *arXiv preprint arXiv:2404.08471*, 2024.

Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeswar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and R Devon Hjelm. Mutual information neural estimation. In *International Conference on Machine Learning (ICML)*, 2018. arXiv:1801.04062.

François Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

Anna Dawid and Yann LeCun. Introduction to latent variable energy-based models: A path towards autonomous machine intelligence. *arXiv preprint arXiv:2306.02572*, 2023. UNVERIFIED arXiv id / authorship – verify before camera-ready.

Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning (ICML)*, 2017. arXiv:1703.03400 – UNVERIFIED id, verify before camera-ready.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2006.07733.

David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.

Danijar Hafner, Kuang-Huei Lee, Ian Fischer, and Pieter Abbeel. Deep hierarchical planning from pixels. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.04114.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. arXiv:2301.04104.

Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.16828.

Yann LeCun. A path towards autonomous machine intelligence. *OpenReview preprint*, 2022. version 0.9.2; OpenReview id BZ5a1r-kVsf.

Yann LeCun, Sumit Chopra, Raia Hadsell, Marc'Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In *Predicting Structured Data*. MIT Press, 2006.

Angsheng Li and Yicheng Pan. Structural information and dynamical complexity of networks. *IEEE Transactions on Information Theory*, 62(6):3290–3339, 2016.

David McAllester and Karl Stratos. Formal limitations on the measurement of mutual information. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020. arXiv:1811.04251.

Meta FAIR. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.

Ofir Nachum, Shixiang Gu, Honglak Lee, and Sergey Levine. Data-efficient hierarchical reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. arXiv:1805.08296.

Ben Poole, Sherjil Ozair, Aaron van den Oord, Alexander A. Alemi, and George Tucker. On variational bounds of mutual information. In *International Conference on Machine Learning (ICML)*, 2019. arXiv:1905.06922.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Alexander Sasha Vezhnevets, Simon Osindero, Tom Schaul, Nicolas Heess, Max Jaderberg, David Silver, and Koray Kavukcuoglu. FeUdal networks for hierarchical reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2017. arXiv:1703.01161.

Weirui Ye, Shaohuai Liu, Thanard Kurutach, Pieter Abbeel, and Yang Gao. Mastering atari games with limited data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2111.00210 – UNVERIFIED id, verify before camera-ready.

Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International Conference on Machine Learning (ICML)*, 2021. arXiv:2103.03230.